

Roles of Mitochondrial DNA in Energy Metabolism.

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Mitochondria are independent double-membrane organelles responsible for energy production, specifically by completing oxidative phosphorylation. Mitochondria are essential to regulate energy metabolism, signaling pathways, and cell death. Mitochondrial DNA (mtDNA) can be altered by metabolic disorders, oxidative stress, or inflammation in the progression and development of various diseases. In this chapter, we overview the role of mtDNA in energy metabolism and the diseases that are associated with mtDNA abnormality, with a special focus on the major factors which regulate the mechanism of mtDNA in metabolism.

The genetics and pathology of mitochondrial disease.

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Mitochondria are double-membrane-bound organelles that are present in all nucleated eukaryotic cells and are responsible for the production of cellular energy in the form of ATP. Mitochondrial function is under dual genetic control - the 16.6-kb mitochondrial genome, with only 37 genes, and the nuclear genome, which encodes the remaining ~1300 proteins of the mitoproteome. Mitochondrial dysfunction can arise because of defects in either mitochondrial DNA or nuclear mitochondrial genes, and can present in childhood or adulthood in association with vast clinical heterogeneity, with symptoms affecting a single organ or tissue, or multisystem involvement. There is no cure for mitochondrial disease for the vast majority of mitochondrial disease patients, and a genetic diagnosis is therefore crucial for genetic counselling and recurrence risk calculation, and can impact on the clinical management of affected patients. Next-generation sequencing strategies are proving pivotal in the discovery of new disease genes and the diagnosis of clinically affected patients; mutations in >250 genes have now been shown to cause mitochondrial disease, and the biochemical, histochemical, immunocytochemical and neuropathological characterization of these patients has led to improved diagnostic testing strategies and novel diagnostic techniques. This review focuses on the current genetic landscape associated with mitochondrial disease, before focusing on advances in studying associated mitochondrial pathology in two, clinically relevant organs - skeletal muscle and brain. © 2016 The Authors. The Journal of Pathology published by John Wiley & Sons Ltd on behalf of Pathological Society of Great Britain and Ireland.

Dysregulated mitochondrial and chloroplast bioenergetics from a translational medical perspective (Review).

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Mitochondria and chloroplasts represent endosymbiotic models of complex organelle development, driven by intense evolutionary pressure to provide exponentially enhanced ATP-dependent energy production functionally linked to cellular respiration and photosynthesis. Within the realm of translational medicine, it has become compellingly evident that mitochondrial dysfunction, resulting in compromised cellular bioenergetics, represents a key causative factor in the etiology and persistence of major diseases afflicting human populations. As a pathophysiological consequence of enhanced oxygen utilization that is functionally uncoupled from the oxidative phosphorylation of ADP, significant levels of reactive oxygen

species (ROS) may be generated within mitochondria and chloroplasts, which may effectively compromise cellular energy production following prolonged stress/inflammatory conditions. Empirically determined homologies in biochemical pathways, and their respective encoding gene sequences between chloroplasts and mitochondria, suggest common origins via entrapped primordial bacterial ancestors. From evolutionary and developmental perspectives, the elucidation of multiple biochemical and molecular relationships responsible for errorless bioenergetics within mitochondrial and plastid complexes will most certainly enhance the depth of translational approaches to ameliorate or even prevent the destructive effects of multiple disease states. The selective choice of discussion points contained within the present review is designed to provide theoretical bases and translational insights into the pathophysiology of human diseases from a perspective of dysregulated mitochondrial bioenergetics with special reference to chloroplast biology.

Assessment of Kidney Mitochondrial Function by High-Resolution Respirometry, Transmission Electron Microscopy, and Histological Techniques.

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Proper kidney function depends highly on mitochondria homeostasis. This organelle is the primary source of ATP production in the kidney and regulates other cellular processes such as redox and calcium homeostasis. Although the mitochondria's primary recognized function is cellular energy production, through the function of the Krebs cycle, electron transport system (ETS), as well as oxygen and electrochemical gradient consumption, this function is interconnected with multiple signaling and metabolic pathways, making bioenergetics a central hub in renal metabolism. Furthermore, mitochondrial biogenesis, dynamics, and mass are also strongly related to bioenergetics. This central role is not surprising given that mitochondrial impairment, including functional and structural alterations, has been recently reported in several kidney diseases. Here, we describe assessment of mitochondrial mass, structure, and bioenergetics in kidney tissue and renal-derived cell lines. These methods allow investigation of mitochondrial alterations in kidney tissue and renal cells under different experimental conditions.

Methodology for measuring oxidative capacity of isolated peroxisomes in the Seahorse assay.

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The regulation of cellular energetics is a complex process that requires the coordinated function of multiple organelles. Historically, studies focused on understanding cellular energy utilization and production have been overwhelmingly concentrated on the mitochondria. While mitochondria account for the majority of intracellular energy production, they alone are incapable of maintaining the variable energetic demands of the cell. The peroxisome has recently emerged as a secondary metabolic organelle that complements and improves mitochondrial performance. Although mitochondria and peroxisomes are structurally distinct organelles, they share key functional similarities that allows for the potential to repurpose readily available tools initially developed for mitochondrial assessment to interrogate peroxisomal metabolic function in a novel manner. To this end, we report here on procedures for the isolation, purification and real-time metabolic assessment of peroxisomal β -oxidation using the Agilent

Seahorse® system. When used together, these protocols provide a straightforward, reproducible and highly quantifiable method for measuring the contributions of peroxisomes to cellular and organismal metabolism.

Lotus germ extract rejuvenates aging fibroblasts via restoration of disrupted proteostasis by the induction of autophagy.

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Cell aging attenuates cellular functions, resulting in time-dependent disruption of cellular homeostasis, which maintains the functions of proteins and organelles. Mitochondria are important organelles responsible for cellular energy production and various metabolic processes, and their dysfunction is strongly related to the progression of cellular aging. Here we demonstrate that disruption of proteostasis attenuates mitochondrial function before the induction of DNA damage signaling by proliferative and replicative cellular aging. We found that lotus (*Nelumbo nucifera* Gaertn.) germ extract clears abnormal proteins and agglutinates via autophagy-mediated restoration of mitochondrial function and cellular aging phenotypes. Pharmacological analyses revealed that DAPK1 expression was suppressed in aging cells, and lotus germ extract upregulated DAPK1 expression by stimulating the acetylation of histones and then induced autophagy by activating the DAPK1-Beclin1 signaling pathway. Furthermore, treatment of aging fibroblasts with lotus germ extract stimulated collagen production and increased contractile ability in three-dimensional cell culture. Thus, time-dependent accumulation of abnormal proteins and agglutinates suppressed mitochondrial function in cells in the early stage of aging, and reactivation of mitochondrial function by restoring proteostasis rejuvenated aging cells. Lotus germ extract rejuvenates aging fibroblasts via the DAPK1-Beclin1 pathway-induced autophagy to clear abnormal proteins and agglutinates.

A simplified model for mitochondrial ATP production.

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Most of the adenosine triphosphate (ATP) synthesized during glucose metabolism is produced in the mitochondria through oxidative phosphorylation. This is a complex reaction powered by the proton gradient across the mitochondrial inner membrane, which is generated by mitochondrial respiration. A detailed model of this reaction, which includes dynamic equations for the key mitochondrial variables, was developed earlier by Magnus and Keizer. However, this model is extraordinarily complicated. We develop a simpler model that captures the behavior of the original model but is easier to use and to understand. We then use it to investigate the mitochondrial responses to glycolytic and calcium input. We use the model to explain experimental observations of the opposite effects of raising cytosolic Ca^{2+} in low and high glucose, and to predict the effects of a mutation in the mitochondrial enzyme nicotinamide nucleotide transhydrogenase (Nnt) in pancreatic beta-cells.

Mitochondrial Genome (mtDNA) Mutations that Generate Reactive Oxygen Species.

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Mitochondria are critical for the energetic demands of virtually every cellular process within nucleated eukaryotic cells. They harbour multiple copies of their own genome (mtDNA), as well as the protein-synthesizing systems required for the translation of vital subunits of the oxidative phosphorylation machinery used to generate adenosine triphosphate (ATP). Molecular lesions to the mtDNA cause severe metabolic diseases and have been proposed to contribute to the progressive nature of common age-related diseases such as cancer, cardiomyopathy, diabetes, and neurodegenerative disorders. As a consequence of playing a central role in cellular energy metabolism, mitochondria produce reactive oxygen species (ROS) as a by-product of respiration. Here we review the evidence that mutations in the mtDNA exacerbate ROS production, contributing to disease.

In situ structure and rotary states of mitochondrial ATP synthase in whole *Polytomella* cells.

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Cells depend on a continuous supply of adenosine triphosphate (ATP), the universal energy currency. In mitochondria, ATP is produced by a series of redox reactions, whereby an electrochemical gradient is established across the inner mitochondrial membrane. The ATP synthase harnesses the energy of the gradient to generate ATP from adenosine diphosphate (ADP) and inorganic phosphate. We determined the structure of ATP synthase within mitochondria of the unicellular flagellate *Polytomella* by electron cryo-tomography and subtomogram averaging at up to 4.2-angstrom resolution, revealing six rotary positions of the central stalk, subclassified into 21 substates of the F1 head. The *Polytomella* ATP synthase forms helical arrays with multiple adjacent rows defining the cristae ridges. The structure of ATP synthase under native operating conditions in the presence of a membrane potential represents a pivotal step toward the analysis of membrane protein complexes in situ.

Cytosolic and Mitochondrial Ca²⁺ Imaging in *Drosophila* Dopaminergic Neurons.

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The ATP-producing organelle mitochondrion controls cellular or synaptic Ca²⁺ concentrations through temporal uptake of Ca²⁺ outside of the mitochondria. Although intracellular Ca²⁺ influx occurs during neuronal activity, a persistently higher concentration of intracellular Ca²⁺ is neurotoxic. Healthy mitochondria ensure rapid Ca²⁺ uptake, which is necessary for proper neuronal activity. Mitochondrial Ca²⁺ buffering activity decreases in aged or sick neurons. In this chapter, we will introduce our protocol for evaluating Ca²⁺ buffering activity through the mitochondria during neuronal activity of dopaminergic neurons.